INCH-POUND
MIL-DTL-24643/82
27 November 2023

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TOPSIDE, MULTI-PAIR, TYPES LSE6U, LSE10U, LSE15U, LSE19U, LSE30U, LSE45U, LSE60U, LSE7AU, LSE19AU, LSE40AU, LSE3TTOP, LSE5TTOP, LSE10TTOP, AND LSE15TTOP (INCLUDING VARIANTS LSE7UW, LSE10UW, LSE19UW, LSE37UW, LSE42UW, LSE7AUW, LSE10AUW, LSE19AUW, LSE37AUW, AND LSE40AUW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LSE**U, LSE**AU, and LSE**TTOP (non-watertight), and LSE**UW and LSE**AUW (watertight).

Construction (Non-Watertight and Watertight)

First	-	Copper conductor, tin-coated, ASTM B286, 7 strands (see table I or II for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Wall thickness as specified in table I or II and as required for components in completed cable to meet the electrical requirements specified herein. Standard identification code applied using method 3.
Fourth	-	Two conductors twisted together to form a pair with a maximum lay length as specified in table I or table II.
Fifth	-	The required number of pairs (see table I or II) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Copper foil/polyester tape. Applied helically with 50 percent minimum overlap (metallic foil side facing out and in contact with the braid).
Eighth	-	Braided shield of tin-coated copper. Braid must be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.
Ninth	-	Copper foil/polyester tape/copper foil. Applied helically with 50 percent minimum overlap. Tape must provide electrical contact between the inner and outer braided shields.

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Tenth - Braided shield of tin-coated copper. Braid must be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be the same as required for the inner braid shield. For watertight constructions only, the use of solid or semi-solid water-block materials (such as silicone, hot melts, or gels) is not allowed on the outer braid. Watertightness shall be by means that could include, but are not limited to, water-swell yarns incorporated within or over the braid or water-swell tape helically applied over the outer braid with 25 percent minimum overlap. The water-block material shall not stick or adhere to the outer braid strands.

Eleventh - An optional binder (tape must not be clear).

Twelfth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.050 inch. See table I and table II for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details (non-watertight).

Military part no. M24643/82	Type and size	Conductor size (AWG)	Number of pairs	Pair lay max. (inches)	Insulation thickness min. avg. (inch)	Cable diameter (inches)	
						min.	max.
-01UO	LSE6U-26	26	6	1.5	0.013	0.445	0.495
-02UO	LSE10U-26	26	10	1.5	0.013	0.515	0.570
-03UO	LSE15U-26	26	15	1.5	0.013	0.560	0.615
-04UO	LSE19U-26	26	19	1.5	0.013	0.585	0.650
-05UO	LSE30U-26	26	30	1.5	0.013	0.695	0.755
-06UO	LSE45U-26	26	45	1.5	0.013	0.825	0.900
-07UO	LSE60U-26	26	60	1.5	0.013	0.905	0.980
-08UO	LSE7AU-22	22	7	2.0	0.020	0.605	0.665
-09UO	LSE19AU-22	22	19	2.0	0.020	0.875	0.950
-10UO	LSE40AU-22	22	40	2.0	0.020	1.210	1.305
-11UO	LSE3TTOP-20	20	3	2.5	0.020	0.515	0.570
-12UO	LSE5TTOP-20	20	5	2.5	0.020	0.595	0.660
-13UO	LSE10TTOP-20	20	10	2.5	0.020	0.735	0.810
-14UO	LSE15TTOP-20	20	15	2.5	0.020	0.830	0.905

TABLE II. Details (watertight).

Military part no. M24643/82	Type and size	Conductor size (AWG)	Number of pairs	Pair lay max. (inches)	Insulation thickness min. avg. (inch)	Cable diameter (inch)	
						min.	max.
-15UO	LSE7UW-26	26	7	1.5	0.013	0.445	0.495
-16UO	LSE10UW-26	26	10	1.5	0.013	0.515	0.570
-17UO	LSE19UW-26	26	19	1.5	0.013	0.585	0.650
-18UO	LSE37UW-26	26	37	1.5	0.013	0.725	0.800
-19UO	LSE42UW-26	26	42	1.5	0.013	0.775	0.850
-20UO	LSE7AUW-22	22	7	2.0	0.020	0.605	0.665
-21UO	LSE10AUW-22	22	10	2.0	0.020	0.720	0.795

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TABLE II. Details (watertight) – Continued.

Military part no. M24643/82	Type and size	Conductor size (AWG)	Number of pairs	Pair lay max. (inches)	Insulation thickness min. avg. (inch)	Cable diameter (inch)	
						min.	max.
-22UO	LSE19AUW-22	22	19	2.0	0.020	0.875	0.950
-23UO	LSE37AUW-22	22	37	2.0	0.020	1.160	1.255
-24UO	LSE40AUW-22	22	40	2.0	0.020	1.210	1.305

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I or II
26 AWG	45.19
22 AWG	17.71
20 AWG	11.53
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	1,500
Conductor to overall shield	1,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to overall shield	500
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Watertightness (applicable to watertight constructions only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	30
Characteristic impedance, at 1 megahertz (ohms)	
26 AWG (non-watertight constructions only)	135±15
26 AWG (watertight constructions only)	115±15
22 AWG and 20 AWG	Not applicable

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Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (applicable to watertight constructions only) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	75
Elongation (percent, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability (applicable to watertight constructions only)	No failure
Shrinkage	No failure
Shield (overall) (conformance to material, construction, and coverage)	No failure
Surface transfer impedance	
Milliohms per meter (max.)	5.0

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Electromagnetic pulse (EMP) response (dB, min.)	80

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (86±3.6 °F [30±2 °C])	No damage
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure
Long term jacket immersion	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All	500

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2017-003)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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